

High-frequency data and limit order books



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U. Paris-Saclay CentraleSupélec cursus Ingénieur 3A Mathématiques et Data Science
IP Paris ENSAE M2 Statistics, Finance and Actuarial Science
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Point processes

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be some probability space.

A point process on $\mathbb{R}_+ = [0, \infty)$ describes the occurrence (times) of random events.

Mathematically, a point process on $\mathbb{R}_+ = [0, \infty)$ can be seen as:

- ▶ a countable random subset of $[0, \infty)$;
- ▶ a sequence of nonnegative random variables ;
- ▶ a counting process ;
- ▶ a discrete random measure.

This lecture focuses on aspects useful for financial modeling. Possible references for further developments outside the scope of this course include [1, 2, 3, 4].

Sequence of random times and counting process

- ▶ Let $0 < T_1 < T_2 < \dots < T_n < \dots$ be an **increasing sequence of random times** of $[0, \infty)$.
- ▶ We assume that T_i 's are distinct (*simple process*) and that $\lim_{n \rightarrow \infty} T_n = +\infty$ (*non-explosive process*).
- ▶ The **counting process** $(N_t)_{t \geq 0}$ associated with this sequence of random times is the stochastic process such that

$$N_t = n \quad \text{if } t \in [T_n, T_{n+1}),$$

or equivalently,

$$N_t = \sup\{n : T_n \leq t\} = \sum_{j \geq 1} \mathbf{1}_{\{T_j \leq t\}}.$$

- ▶ The random variables $S_n = T_n - T_{n-1}$ are called **interarrival times**, or **durations**.

Properties of a counting process

We thus have the following properties:

- (i) $N_0 = 0$;
- (ii) a counting process is **non-negative integer-valued** ;
- (iii) a counting process is **non-decreasing** ;
- (iv) a counting process is **piecewise constant** ;
- (v) a counting process is **càdlàg** (right continuous with left limits) ;
- (vi) all jumps of a counting process are of **size 1**.

The following relationships are often useful:

- ▶ $\{N_t \geq n\} = \{T_n \leq t\}$,
- ▶ $\{N_t = n\} = \{T_n \leq t < T_{n+1}\}$,
- ▶ $\{N_t < n \leq N_{t+s}\} = \{t < T_n \leq t + s\}$.

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Construction with interarrival times I

- ▶ In this section, we review the standard construction of the Poisson process on the real half-line.
- ▶ Let $(S_i)_{i \in \mathbb{N}^*}$ be a sequence of **independent and identically distributed** random variables, with **exponential distribution** with parameter $\lambda \in \mathbb{R}_+^*$.
- ▶ Let $T_n = \sum_{i=1}^n S_i$, $n \geq 1$. $(T_n)_{n \geq 1}$ is a point process on $[0, +\infty)$.
- ▶ Let $N_t = \sum_{j \geq 1} \mathbf{1}_{\{T_j \leq t\}}$ be the associated counting process.

Proposition (Distributional properties)

- (a) For any $n \in \mathbb{N}^*$, T_n is **Gamma distributed** with shape parameter n and rate parameter λ , i.e. with density $x \mapsto \frac{(\lambda x)^{n-1}}{(n-1)!} \lambda e^{-\lambda x} \mathbf{1}_{\mathbb{R}_+}(x)$.
- (b) For any $t \in \mathbb{R}_+$, N_t is **Poisson distributed** with parameter λt , i.e.
- $$\mathbf{P}(N_t = k) = \frac{(\lambda t)^k}{k!} e^{-\lambda t}, k \in \mathbb{N}.$$

Construction with interarrival times II

Sketch of the proofs.

- (a) $T_1 = S_1$, so $f_{T_1}(x) = \lambda e^{-\lambda x} \mathbf{1}_{\mathbb{R}_+}(x)$ and the property is valid for $n = 1$. $T_{n+1} = T_n + S_{n+1}$ and T_n and S_{n+1} are independent, so densities satisfy

$$f_{T_{n+1}}(x) = f_{T_n + S_{n+1}}(x) = (f_{T_n} * f_{S_{n+1}})(x) = \int_{\mathbb{R}} f_{T_n}(u) f_{S_{n+1}}(x - u) du.$$

If $f_{T_n}(x) = \frac{(\lambda x)^{n-1}}{(n-1)!} \lambda e^{-\lambda x} \mathbf{1}_{\mathbb{R}_+}(x)$, then

$$\begin{aligned} f_{T_{n+1}}(x) &= \int_{\mathbb{R}_+} \frac{(\lambda u)^{n-1}}{(n-1)!} \lambda e^{-\lambda u} \lambda e^{-\lambda(x-u)} \mathbf{1}_{\{x-u \geq 0\}} du = \lambda^{n+1} e^{-\lambda x} \mathbf{1}_{\mathbb{R}_+}(x) \int_0^x \frac{u^{n-1}}{(n-1)!} du \\ &= \lambda e^{-\lambda(x)} \mathbf{1}_{\mathbb{R}_+}(x) \frac{(\lambda x)^n}{n!}, \end{aligned}$$

which proves the result by induction.

Construction with interarrival times III

(b) $\mathbf{P}(N_t = n) = \mathbf{P}(T_n \leq t) - \mathbf{P}(T_{n+1} \leq t)$ and integrating by parts:

$$\begin{aligned}\mathbf{P}(T_{n+1} \leq t) &= \int_0^t \frac{(\lambda x)^n}{n!} \lambda e^{-\lambda x} dx \\ &= \left[-\frac{(\lambda x)^n}{n!} e^{-\lambda x} \right]_0^t + \int_0^t \frac{(\lambda x)^{n-1}}{(n-1)!} \lambda e^{-\lambda x} dx \\ &= -e^{-\lambda t} \frac{(\lambda t)^n}{n!} + \mathbf{P}(T_n \leq t),\end{aligned}$$

hence the result.

Definition following the construction

Definition (Time-homogeneous Poisson process)

A process $(N_t)_{t \geq 0}$ is called a **time-homogeneous Poisson process** with parameter $\lambda > 0$ if it is the counting process associated to a point process with **i.i.d. interarrival times, exponentially distributed** with parameter $\lambda > 0$.

- Remark : Construction can be generalized to i.i.d. interarrival times with any distribution (with finite mean). Such processes are called *renewal processes*. But without the memorylessness property of the exponential distribution, non-Poisson renewal processes do not have stationary and independent increments (see next slides), or satisfy the Markov property.

Increments of a Poisson process I

Proposition (Increments)

- (a) *Increments of the process $(N_t)_{t \geq 0}$ are independent, i.e. for all $n \in \mathbb{N}^*$ and all strictly increasing finite sequence $0 < t_1 < \dots < t_n$, the random variables $N_{t_1}, N_{t_2} - N_{t_1}, \dots, N_{t_n} - N_{t_{n-1}}$ are mutually independent.*
- (b) *Increments of the process $(N_t)_{t \geq 0}$ are stationary, i.e. for any $s < t$, $N_t - N_s$ and N_{t-s} have the same distribution, which is Poisson with parameter $\lambda(t - s)$.*

Increments of a Poisson process II

Sketch of the proofs. Let $\mathcal{F}_t = \sigma(\{N_s : 0 \leq s \leq t\})$. These properties are the consequence of the **memorylessness property** of the exponential distribution: if $X \sim \mathcal{E}(\lambda)$, then

$$\mathbf{P}(X > t + h \mid X > t) = \frac{\mathbf{P}(X > t + h)}{\mathbf{P}(X > t)} = \frac{e^{-\lambda(t+h)}}{e^{-\lambda t}} = \mathbf{P}(X > h).$$

Thus, at any time s , the distribution of the duration to the next jump is independent of $\mathcal{F}_s$, and the argument is valid for all subsequent jumps. Hence for all $t > s$, $N_t - N_s$ is independent of $\mathcal{F}_s$. (a) is then obtained by induction on t_i 's. The same argument shows that for all $t > s$, $N_t - N_s$ has the same distribution as N_{t-s} , which is (b).

Characterization by the increments

- Independent and stationary increments in fact characterize the Poisson process.

Lemma (Memoryless distribution)

A non-negative (and not identically zero) random variable has the memorylessness property if and only if it has an exponential distribution.

Proposition (Poisson characterization by the increments)

Let N be a (simple, locally integrable, and not identically zero) counting process. Then N is a Poisson process if and only if N has stationary and independent increments.

An alternative definition

- ▶ The following alternative definition is a consequence of the previous characterization.

Definition (Time-homogeneous Poisson process)

A counting process $(N_t)_{t \geq 0}$ is called a **time-homogeneous Poisson process** with parameter $\lambda > 0$ if

- (i) **N has independent increments**: for all $k \in \mathbb{N}^*$, for any collection $I_1, \dots, I_k$ of pairwise disjoint intervals of $[0, \infty)$, the random variables $N(I_1), \dots, N(I_k)$ are independent ;
- (ii) the number of events in any interval with length s is Poisson distributed with parameter λs :

$$\forall s, t \geq 0, \forall n \in \mathbb{N}, \mathbf{P}(N_{t+s} - N_t = n) = e^{-\lambda s} \frac{(\lambda s)^n}{n!}.$$

- ▶ This form is important since this is a specific case of a general definition of Poisson processes defined by random measures on arbitrary measurable spaces (e.g., $\mathbb{R}^d$ for space modeling purposes), see next section.

Asymptotic behavior of a Poisson process - Law of large numbers

Proposition (Law of large numbers)

If $(N_t)_{t \geq 0}$ is a (time-homogeneous) Poisson process with parameter $\lambda > 0$, then

$$\lim_{t \rightarrow \infty} \frac{N_t}{t} = \lambda, \quad \mathbf{P}\text{-a.s. and in } L^2(\Omega, \mathcal{F}, \mathbf{P}).$$

Sketch of the proof. $\mathbf{E} \left[\frac{N_t}{t} \right] = \lambda$, hence $\mathbf{E} \left[\left| \frac{N_t}{t} - \lambda \right|^2 \right] = \mathbf{V} \left[\frac{N_t}{t} \right] = \frac{\lambda}{t} \xrightarrow{t \rightarrow +\infty} 0$, which proves the L^2 convergence. Moreover, for any $n \in \mathbb{N}^*$,

$$\frac{N_n}{n} = \frac{1}{n} \sum_{i=1}^n (N_i - N_{i-1}) \xrightarrow{n \rightarrow +\infty} \mathbf{E}[(N_1 - N_0)] = \lambda \quad \text{a.s.}$$

by the strong law of large numbers. Finally, for any $t \in \mathbb{R}_+$, $\lfloor t \rfloor \leq t < \lfloor t \rfloor + 1$ yields

$$\frac{N_{\lfloor t \rfloor}}{\lfloor t \rfloor} \frac{\lfloor t \rfloor}{t} \leq \frac{N_t}{t} < \frac{N_{\lfloor t \rfloor + 1}}{\lfloor t \rfloor + 1} \frac{\lfloor t \rfloor + 1}{t},$$

hence the result.

Asymptotic behavior of a Poisson process - Central limit theorem I

Proposition (Central limit theorem)

If $(N_t)_{t \geq 0}$ is a (time-homogeneous) Poisson process with parameter $\lambda > 0$, then

$$\frac{N_t - \lambda t}{\sqrt{\lambda t}} \xrightarrow[t \rightarrow \infty]{\mathcal{L}} Z,$$

where Z is a standard Gaussian random variable.

Sketch of the proof. For any $n \in \mathbb{N}^*$,

$$\frac{N_n - n\lambda}{\sqrt{n\lambda}} = \frac{\sum_{i=1}^n (N_i - N_{i-1}) - n\lambda}{\sqrt{n\lambda}} \xrightarrow[n \rightarrow +\infty]{} Z \sim \mathcal{N}(0, 1) \quad \text{in distribution}$$

by the central limit theorem. In the general case, for any $t \in \mathbb{R}_+$, we write

$$\frac{N_t - \lambda t}{\sqrt{\lambda t}} = \frac{N_t - N_{\lfloor t \rfloor}}{\sqrt{\lambda \lfloor t \rfloor}} \frac{\sqrt{\lambda \lfloor t \rfloor}}{\sqrt{\lambda t}} + \frac{N_{\lfloor t \rfloor} - \lambda \lfloor t \rfloor}{\sqrt{\lambda \lfloor t \rfloor}} \frac{\sqrt{\lambda \lfloor t \rfloor}}{\sqrt{\lambda t}} + \frac{\lambda \lfloor t \rfloor - \lambda t}{\sqrt{\lambda \lfloor t \rfloor}} \frac{\sqrt{\lambda \lfloor t \rfloor}}{\sqrt{\lambda t}}.$$

Asymptotic behavior of a Poisson process - Central limit theorem II

The second term converges in distribution to the expected limit and the third term converges to 0. For the first term, $N_t \leq N_{\lfloor t \rfloor + 1}$ implies

$$\frac{N_t - N_{\lfloor t \rfloor}}{\sqrt{\lambda \lfloor t \rfloor}} \leq \frac{N_{\lfloor t \rfloor + 1} - N_{\lfloor t \rfloor}}{\sqrt{\lambda \lfloor t \rfloor}} := \xi_{\lfloor t \rfloor}.$$

Then for $\epsilon > 0$,

$$\mathbf{P}(\xi_n > \epsilon) = \mathbf{P}(N_{n+1} - N_n > \epsilon\sqrt{\lambda n}) = \mathbf{P}(\mathcal{P}(\lambda) > \epsilon\sqrt{\lambda n}) \xrightarrow{n \rightarrow +\infty} 0,$$

and the first term converges in probability towards 0. The result is thus obtained using Slutsky's theorem.

A functional Central Limit Theorem

- ▶ Let $(N_t)_{t \geq 0}$ be a (time-homogeneous) Poisson process with parameter $\lambda > 0$. For any $t, u \geq 0$, let us set

$$B_t^{(u)} = \frac{N_{tu} - \lambda tu}{\sqrt{\lambda u}}.$$

For any given $u \geq 0$, $(B_t^{(u)})_{t \geq 0}$ is a stochastic process with independent increments, and jump sizes $\frac{1}{\sqrt{\lambda u}}$.

- ▶ Given the previous result, for all $t \geq 0$,

$$B_t^{(u)} = \frac{N_{tu} - \lambda tu}{\sqrt{\lambda u}} \xrightarrow[u \rightarrow \infty]{\mathcal{L}} Z_t,$$

where Z_t is a Gaussian random variable with mean 0 and variance t .

- ▶ When $u \rightarrow +\infty$, we therefore obtain as a limit a Gaussian process, with independent increments, with continuous paths, with mean 0 and variance t , i.e. a **Brownian motion**.

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Random counting measure I

- ▶ In this section, we temporarily leave the real-half line $[0, \infty)$ for a space E a bit more general (typically, $\mathbb{R}^d$).
- ▶ Let $(\Omega, \mathcal{F}, \mathbf{P})$ be some probability space and $(E, \mathcal{B}(E))$ be a measurable space.
- ▶ A **random measure** on a $(E, \mathcal{B}(E))$ is a mapping

$$N : \Omega \times \mathcal{B}(E) \rightarrow [0, \infty],$$

such that

- (i) for a given $\omega \in \Omega$, $A \mapsto N(\omega, A)$ is a measure on $(E, \mathcal{B}(E))$;
 - (ii) for a given $A \in \mathcal{B}(E)$, $\omega \mapsto N(\omega, A)$ is a random variable.
- ▶ We very often write $N(A)$ instead of $N(\omega, A)$.

Random counting measure II

- ▶ A **point process** on E is a countable collection $(X_i)_{i \geq 1}$ of random points of E .
- ▶ We assume that it is locally finite, i.e. for all bounded set $B \in \mathcal{E}$,
 $\#\{i \in \mathbb{N}^* : X_i \in B\} < \infty$ a.s.
- ▶ This defines a **random, discrete, counting measure** $N = \sum_{i \geq 1} \delta_{X_i}$ on $(E, \mathcal{E})$ by setting for all

$A \in \mathcal{B}(E)$:

$$N(A) = \#\{i \in \mathbb{N}^* : X_i \in A\} = \sum_{i \geq 1} \mathbf{1}_A(X_i) = \sum_{i \geq 1} \delta_{X_i}(A).$$

- ▶ For any non-negative measurable function $f : E \rightarrow \mathbb{R}_+$, we write

$$\int_E f(x) N(dx) = \sum_{i \geq 1} f(X_i).$$

Intensity measure of a point process

Proposition (Intensity measure of a point process)

The *intensity measure* of a random measure N is the (deterministic) measure μ on $(E, \mathcal{E})$ defined for all $A \in \mathcal{B}(E)$ by:

$$\mu(A) = \mathbf{E}[N(A)]$$

- ▶ μ is indeed a measure.
- ▶ For a non-negative measurable function $f : E \rightarrow \mathbb{R}_+$, we have the **Campbell's formula**:

$$\mathbf{E} \left[\int_E f(x) N(dx) \right] = \int_E f(x) \mu(dx).$$

Sketch of the proof. First for indicator functions $\mathbf{1}_A$, then for simple functions by linearity, then for non-negative measurable functions by monotone convergence.

Poisson random measures

Definition (Poisson random measures)

A random counting measure $N = \sum_{i \geq 1} \delta_{X_i}$ is called a **Poisson random measure** (or Poisson point process) **with intensity measure μ** if:

- (i) for any finite collection of pairwise disjoint sets $A_1, \dots, A_n \in \mathcal{B}(E)$, the random variables $N(A_1), \dots, N(A_n)$ are mutually independent ;
- (ii) for any bounded set $A \in \mathcal{B}(E)$, $\mathbf{P}(N(A) = k) = \frac{(\mu(A))^k e^{-\mu(A)}}{k!}$, $k = 0, 1, 2, \dots$

- ▶ We have $\mathbf{E}[N(A)] = \mu(A)$ so μ is indeed the intensity measure of N .
- ▶ When $E = [0, +\infty)$ and $\mu(A) = \lambda \ell(A)$ for some $\lambda > 0$, where ℓ is the Lebesgue measure, this definition coincides with the above definition of the time-homogeneous Poisson process with intensity (parameter/function) $\lambda > 0$.
- ▶ For the existence in the general case, see, e.g., [4].

Superposition theorem

Theorem (Superposition theorem)

Let $n \geq 2$. Let $N_1, N_2, \dots, N_n$ be independent Poisson random measures on E with intensity measures $\mu_1, \mu_2, \dots, \mu_n$. Then the sum $N = \sum_{k=1}^n N_k$ is a Poisson random measure with intensity $\mu = \sum_{k=1}^n \mu_k$.

Sketch of the proof:

- (i) For any bounded set $A \in \mathcal{B}(E)$, $N_k(A)$ is a Poisson random variable with parameter $\mu_k(A)$ so $N(A) = \sum_{k=1}^n N_k(A)$ is a Poisson random variable with parameter $\mu(A) = \sum_{k=1}^n \mu_k(A)$ (sum of independent Poisson random variables).
- (ii) For any finite collection of pairwise disjoint sets $A_1, \dots, A_m \in \mathcal{B}(E)$, $\{N_k(A_i) : k = 1, \dots, n, i = 1, \dots, m\}$ is a set of mutually independent variables, so that, by grouping, the random variables $N(A_i) = \sum_{k=1}^n N_k(A_i)$, $i = 1, \dots, m$ are independent random variables.

Conditional distributions I

Lemma (Conditional distributions on subsets)

Let N be a Poisson random measure on $(E, \mathcal{B}(E))$ with intensity measure μ . Let $A \in \mathcal{B}(E)$ such that $\mu(A) < +\infty$. Let $(B_0, B_1, \dots, B_k)$ a partition of A . Then

$$(N(B_0), N(B_1), \dots, N(B_k)) | N(A) = n$$

has a *multinomial distribution* with parameters $\left(n; \frac{\mu(B_0)}{\mu(A)}, \dots, \frac{\mu(B_k)}{\mu(A)}\right)$.

Sketch of the proof: $N(B_0), \dots, N(B_k)$ are independent Poisson r.v. with parameters $\mu(B_0), \dots, \mu(B_k)$. Let $n_i \in \mathbb{N}$, $i = 0, \dots, k$ such that $\sum_{i=0}^k n_i = n$. Then by direct computation

$$\mathbf{P}\left(\bigcap_{i=0}^k N(B_i) = n_i | N(A) = n\right) = \frac{n!}{\prod_{i=0}^k n_i!} \prod_{i=0}^k \left(\frac{\mu(B_i)}{\mu(A)}\right)^{n_i}.$$

Conditional distributions II

- ▶ **Reminder** : The multinomial distribution with parameters $(n; p_0, \dots, p_n)$, where $\sum_{i=0}^n p_i = 1$, counts the number of occurrences of n i.i.d. trials falling into category $i = 0, \dots, n$ with probability $p_0, \dots, p_n$.
- ▶ **Consequence** : given $N(A) = n$, the points in A are independent and identically distributed, i.e. if $(X_1, \dots, X_{N(A)})$ are the points of N in A , then the random variables

$$X_1 | N(A) = n, \dots, X_n | N(A) = n$$

are i.i.d. with distribution $\mu(B) = \frac{\mu(B \cap A)}{\mu(A)}$, $B \in \mathcal{B}(E)$.

Marking theorem I

Theorem ((Discrete) Marking theorem)

Let N be a Poisson random measure on E with intensity μ . For each point X_i in N , we independently assign a mark M_i in $\{1, \dots, M\}$ according to a probability distribution $\mathbf{P}(M_i = m | X_i = x) = p_m(x)$. Let $N^{(m)}$ be the random measure of points with mark m , i.e.

$$\forall A \in \mathcal{B}(E), N^{(m)}(A) = \sum_{i \geq 1} \delta_{X_i}(A) \mathbf{1}_{M_i=m}.$$

Then the processes $N^{(m)}$ for different marks m are *independent Poisson random measures* with intensities defined for $A \in \mathcal{B}(E)$ by $\mu^{(m)}(A) = \int_A p_m(x) \mu(dx)$.

- ▶ Discrete mark case for simplicity, covers most/all practical uses
- ▶ The mark M_i may depend on X_i , but not on other points or marks.

Marking theorem II

Sketch of the proof. Let $q_m(A) = \frac{1}{\mu(A)} \int_A p_m(x) \mu(dx)$. Computation gives for $t \in \mathbb{R}$ the probability generating function,

$$\mathbf{E}[t^{N^{(m)}(A)} | N(A) = n] = (1 - q_m(A) + q_m(A)t)^n,$$

i.e. $N^{(m)}(A) | N(A) = n$ has a binomial distribution with parameters $(n, q_m(A))$. Then by summing over n to remove the condition $N(A) = n$, we get

$$\mathbf{P}(N^{(m)}(A) = k) = e^{-\mu(A)q_m(A)} \frac{(\mu(A)q_m(A))^k}{k!},$$

i.e. $N^{(m)}(A)$ is a Poisson r.v. with parameter $\mu(A)q_m(A)$. By repeating these computations for all A_j where $A_1, \dots, A_n$ are disjoint sets, one obtains that $N^{(m)}(A_j)$ are independent Poisson r.v. and conclude.

Marking theorem III

Corollary (Restriction)

The restriction of a Poisson random measure N with intensity measure μ to $A \in \mathcal{B}(E)$ is a Poisson random measure with intensity $\mu(\cdot \cap A)$.

Corollary (Thinning theorem for Poisson random measures)

Let N be a Poisson random measure on $(E, \mathcal{B}(E))$ with intensity measure μ . Let $p : E \rightarrow [0, 1]$ be a measurable function. Suppose each point X_i of N is removed with probability $1 - p(X_i)$. Then the set $\tilde{N}$ of kept points is a Poisson random measure on $(E, \mathcal{B}(E))$ with intensity measure $\tilde{\mu}$ defined by

$$\tilde{\mu}(B) = \int_B p(x) \mu(dx)$$

Back to the real half-line

- ▶ Let us go back to the case $E = [0, +\infty)$.
- ▶ If $\mu(A) = \lambda \ell(A)$ for some $\lambda > 0$, where ℓ is the Lebesgue measure, then the definition of the Poisson random measure with intensity measure μ coincides with the definition of the time-homogeneous Poisson process with intensity $\lambda > 0$.
- ▶ Now, let $\lambda : [0, \infty) \rightarrow \mathbb{R}_+$ be a locally integrable function. Let $\Lambda : \mathcal{B}([0, \infty)) \rightarrow \mathbb{R}_+$ be the measure on $[0, +\infty)$ defined by

$$\Lambda(A) = \int_A \lambda(s) ds.$$

A Poisson random measure with intensity measure Λ is called a non-homogeneous Poisson process with intensity (function) λ .

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(Non-homogeneous) Poisson processes I

Definition (Non-homogeneous Poisson process)

Let $\lambda : [0, \infty) \rightarrow \mathbb{R}_+$ be a locally integrable function. Let

$$\Lambda(t) = \int_0^t \lambda(s) ds, \quad t \geq 0.$$

A counting process N is called a **Poisson process with intensity λ** if

- (i) N has independent increments ;
- (ii) for all $0 \leq s \leq t$, the random variable $N_t - N_s$ is Poisson distributed with parameter $\Lambda(t) - \Lambda(s)$.

- ▶ Recall that we have assumed $N_0 = 0$ for all counting processes.
- ▶ λ is a **deterministic** function.

(Non-homogeneous) Poisson processes II

- Increments are **not stationary** anymore (except of course in the constant λ case).
- Of course, $\mathbf{E}[N_t] = \Lambda(t)$ and $\mathbf{V}[N_t] = \Lambda(t)$ (Poisson distribution).
- Then we have the **local Poisson behaviour** (a.e.):

$$\begin{cases} \mathbf{P}(N_{t+h} - N_t = 0) = 1 - \lambda(t)h + o(h), \\ \mathbf{P}(N_{t+h} - N_t = 1) = \lambda(t)h + o(h), \\ \mathbf{P}(N_{t+h} - N_t \geq 2) = o(h). \end{cases}$$

Sketch of the proof. $N_{t+h} - N_t$ is Poisson distributed with parameter $\int_t^{t+h} \lambda(s) ds = \Lambda(t+h) - \Lambda(t) = \lambda(t)h + o(h)$ a.e., so

$$\mathbf{P}(N_{t+h} - N_t = 0) = \exp\left(-\int_t^{t+h} \lambda(s) ds\right) = 1 - \lambda(t)h + o(h).$$

Similar computations for the other cases.

Time change for a Poisson process I

Proposition (Time change for a Poisson process)

Let $\lambda : \mathbb{R}_+ \rightarrow \mathbb{R}_+^*$ be a locally integrable function and for all $t \geq 0$, let $\Lambda(t) = \int_0^t \lambda(s) ds$. Let N be a time-homogeneous Poisson process with constant intensity 1. Then $\tilde{N}_t = N_{\Lambda(t)}$ is a *non-homogeneous Poisson process* with intensity $t \mapsto \lambda(t)$.

Sketch of the proof: Let $(T_n)_n$ be the jump times of process N , and let

$$S_n = \Lambda^{-1}(T_n) = \inf\{t \geq 0 : \Lambda(t) \geq T_n\}.$$

Then

$$\tilde{N}_t = N_{\Lambda(t)} = \sum_{i \geq 1} \mathbf{1}_{T_i \leq \Lambda(t)} = \sum_{i \geq 1} \mathbf{1}_{\Lambda^{-1}(T_i) \leq t} = \sum_{i \geq 1} \mathbf{1}_{S_i \leq t},$$

i.e. $\tilde{N}$ is the counting process associated to the event times $(S_i)_i$. Furthermore, for any $0 < t_1 < \dots < t_n$, $\tilde{N}_{t_i} - \tilde{N}_{t_{i-1}} = N_{\Lambda(t_i)} - N_{\Lambda(t_{i-1})}$, hence increments are independent and Poisson distributed.

Time change for a Poisson process II

Proposition (Integrated intensity between event times)

Let $\tilde{N}$ be a non-homogeneous Poisson process with intensity $t \mapsto \lambda(t)$. Let $(S_i)_{i \geq 1}$ be the random times of counted events. Then the random variables $(\Lambda(S_i) - \Lambda(S_{i-1}))_{i \geq 2}$ are *i.i.d.* random variables, *exponentially distributed* with parameter 1.

Sketch of the proof: Using the above construction,

$$\Lambda(S_i) - \Lambda(S_{i-1}) = \Lambda(\Lambda^{-1}(T_i)) - \Lambda(\Lambda^{-1}(T_{i-1})) = T_i - T_{i-1} \sim \mathcal{E}(1).$$

- This can be used as a goodness-of-fit test of non-homogeneous Poisson processes to empirical data.

Thinning for non-homogeneous Poisson processes

Theorem (Thinning for Poisson processes on $[0, \infty)$)

Let N^* be a (non-homogeneous) Poisson process with intensity $\lambda^*(t)$, $t \in [0, \infty)$. Let $\lambda(t)$ be a deterministic positive function *such that* $\lambda(t) \leq \lambda^*(t)$, $t \geq 0$. Let $T_1^*, \dots, T_{N_T^*}^*$ be the sequence of random points of N^* on $[0, T]$. For $i = 1, \dots, N_T^*$, *delete the point* T_i^* *with probability* $1 - \frac{\lambda(T_i^*)}{\lambda^*(T_i^*)}$. Then the remaining points form a (non-homogeneous) Poisson process with intensity $\lambda(t)$ on $[0, T]$.

- ▶ This is a direct consequence of the thinning theorem for Poisson random measures.
- ▶ Algorithm introduced in [5].

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Time-homogeneous Poisson simulation

Given the results of the previous sections, we have immediately two simulation algorithms for a time-homogeneous Poisson process with parameter $\lambda > 0$ on $[0, T]$, for some horizon $T > 0$.

► Simulation using interarrival times

1. $t = 0, i = 1$
2. Generate $s \sim \mathcal{E}(\lambda)$ (i.e. generate $u \sim \mathcal{U}([0, 1])$ and set $s = -\frac{1}{\lambda} \log(u)$).
3. If $t + s \leq T$:
 - 3.1 Set $t_i = t + s, t = t + s, i = i + 1$
 - 3.2 Go to 2.
4. Else: Return $(t_i)_{i \geq 1}$

► Simulation using order statistics

1. Generate $n \sim \mathcal{P}(\lambda T)$
2. Generate $u_i \sim \mathcal{U}([0, T]), i = 1, \dots, n$, i.i.d.
3. Return $(t_i)_{i=1, \dots, n}$ the ordered sequence of $\{u_i, i = 1, \dots, n\}$.

Non-homogeneous Poisson simulation

Simulation of a non homogeneous Poisson process on $[0, T]$, for some horizon $T > 0$, with deterministic intensity function $\lambda : [0, T] \rightarrow \mathbb{R}_+$, and integrated intensity $\Lambda : t \mapsto \int_0^t \lambda(s) ds$.

- **Simulation using interval times** We mimic the time-homogeneous case, except interarrival times are not exponentially distributed, but should be generated from

$$\begin{aligned} F(s) &= \mathbf{P}(T_n \leq t_{n-1} + s | T_{n-1} = t_{n-1}) = 1 - \mathbf{P}(T_n > t_{n-1} + s | T_{n-1} = t_{n-1}) \\ &= 1 - \mathbf{P}(N_{t_{n-1}+s} - N_{t_{n-1}} = 0 | T_{n-1} = t_{n-1}) = 1 - \mathbf{P}(N_{t_{n-1}+s} - N_{t_{n-1}} = 0) \\ &= 1 - \exp(-(\Lambda(t_{n-1} + s) - \Lambda(t_{n-1}))). \end{aligned}$$

(Need for an efficient inversion of F)

- **Simulation using time change**
 1. Generate $(t_i)_{i=1,\dots,n}$ sample path of a time-homogeneous Poisson process with intensity 1.
 2. Return $(\Lambda^{-1}(t_i))_{i=1,\dots,n}$.

(Need for an efficient inversion of Λ)

NHPP simulation by thinning

The previous simulations often fail to be either numerically efficient or practical. **Thinning** is a flexible method to simulate non-homogeneous Poisson processes (and more, see later) if for example λ is bounded by a constant $\lambda^* \in \mathbb{R}_+^*$ on $[0, T]$.

► Global thinning

1. Generate $(T_i^*)_{i=1,\dots,n}$ a realization of a time-homogeneous Poisson process with parameter λ^* .
2. Generate $u_i \sim \mathcal{U}([0, 1])$, $i = 1, \dots, n$.
3. Return the set of points $\left\{ T_i^* : u_i \leq \frac{\lambda(T_i^*)}{\lambda^*} \right\}$.

► Thinning per interarrival time

1. $t = 0, i = 1$
2. Generate $s \sim \mathcal{E}(\lambda^*)$ (i.e. $s = -\frac{1}{\lambda^*} \log(u)$, $u \sim \mathcal{U}([0, 1])$).
3. If $t + s \leq T$:
 - 3.1 Generate $u \sim \mathcal{U}[0, 1]$
 - 3.2 If $u \leq \frac{\lambda(t+s)}{\lambda^*}$: Set $t_i = t + s$, $i = i + 1$
 - 3.3 $t = t + s$
 - 3.4 Go to 2.
4. Else: Return $(t_i)_{i \geq 1}$

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Lab 2 - Questions I

1. Non-homogeneous Poisson process and trade times.

- 1.1 Implement a thinning algorithm for the simulation of a non-homogeneous Poisson process on $[0, T]$ with a given intensity $\mu : [0, T] \rightarrow \mathbb{R}_+$. Make some tests to check that your simulation algorithm is correct.
- 1.2 For each stock in your dataset, propose/calibrate a deterministic intensity function f that represents the intensity of occurrence of trades during a trading day. Using simulations of a (non-homogeneous) Poisson process with intensity f , comment on the statistical properties of the fitted model and on the goodness-of-fit of this model.

Lab 2 - Questions II

2. **Brownian motions with Poisson sampling.** Consider two processes $p_i(t) = \sigma_i W_i(t)$, $i = 1, 2$, where W_1 and W_2 are two Brownian motions such that $\langle W_1, W_2 \rangle_t = \rho t$ for some constants $\sigma_1 > 0$, $\sigma_2 > 0$ and $\rho \in]-1, 1[$. Assume that for $i = 1, 2$ the process p_i is observed on $[0, T]$ at times $T_{i,k}$, $k \geq 0$, and that $(T_{i,k})_{k \geq 0}$ is a Poisson process with constant intensity $\mu_i > 0$. Such a framework could be used to model the trade times and log-trade prices for two correlated assets. Simulate multiple sample paths of the *observed* processes (trade times and prices) and on each path compute the standard covariance estimator of $[p_1, p_2]_T = \rho \sigma_1 \sigma_2 T$ at various sampling periods τ . Comment.

Lab 2 - Questions III

3. **Empirical intensities and LOB features.** The scientific literature on LOB suggests that intensities of order flows depend on observed LOB features. We will discuss this later in the course. On your dataset, measure (some proxy of) the empirical intensities of trades as a function
- 3.1 of the (rescaled) observed queue size ;
 - 3.2 of the observed spread.

Explain precisely your method and comment in both cases.

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